

ENV 6106

Air Pollution Modeling

Spring 2023

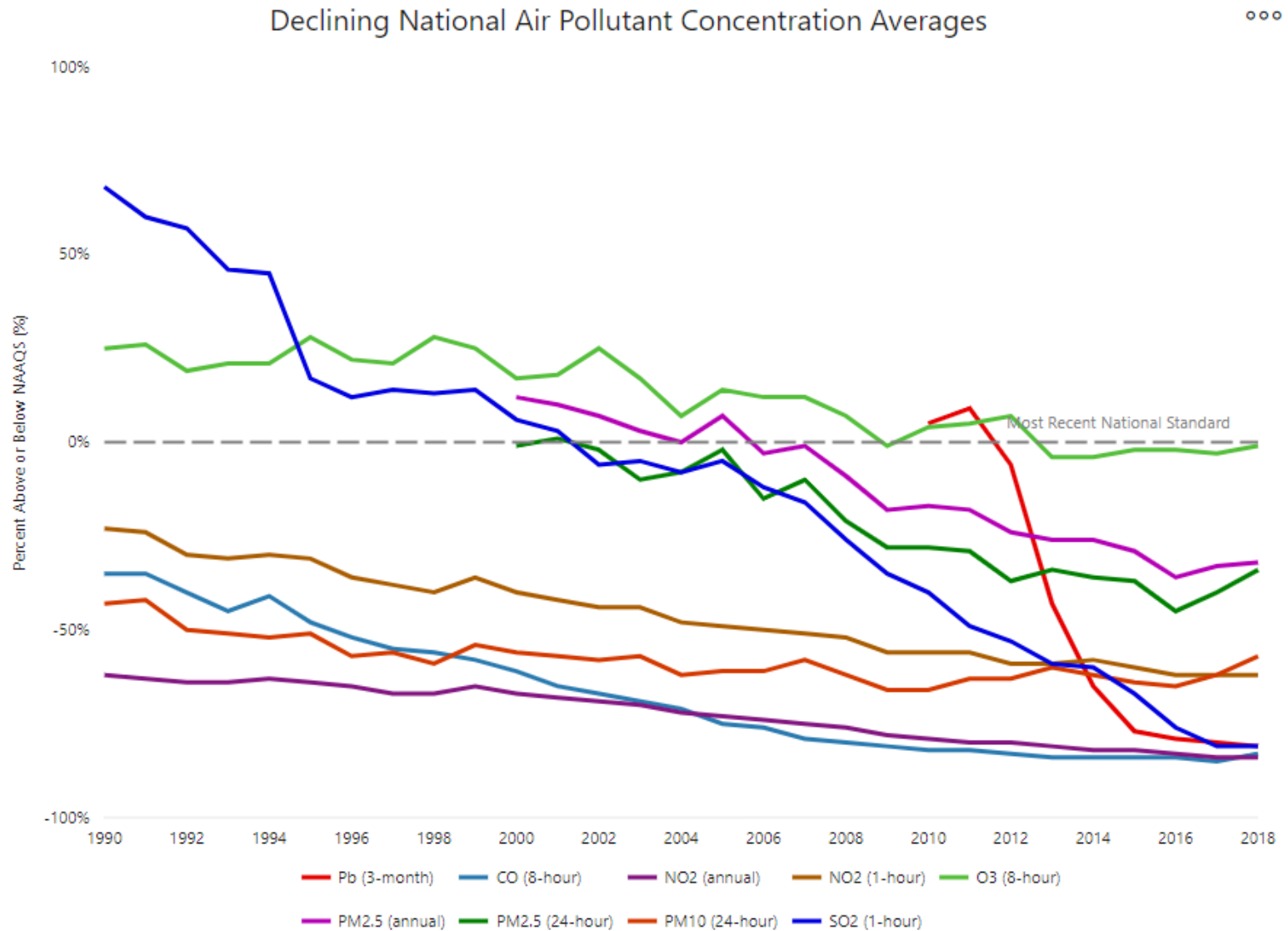
Topics

- Overview of mobile source and intro to MOVES
- Expectation
 - Understand the importance of mobile source emissions
 - Understand what MOVES model
- Reference:
 - MOVES user guide:

CAPs & HAPs

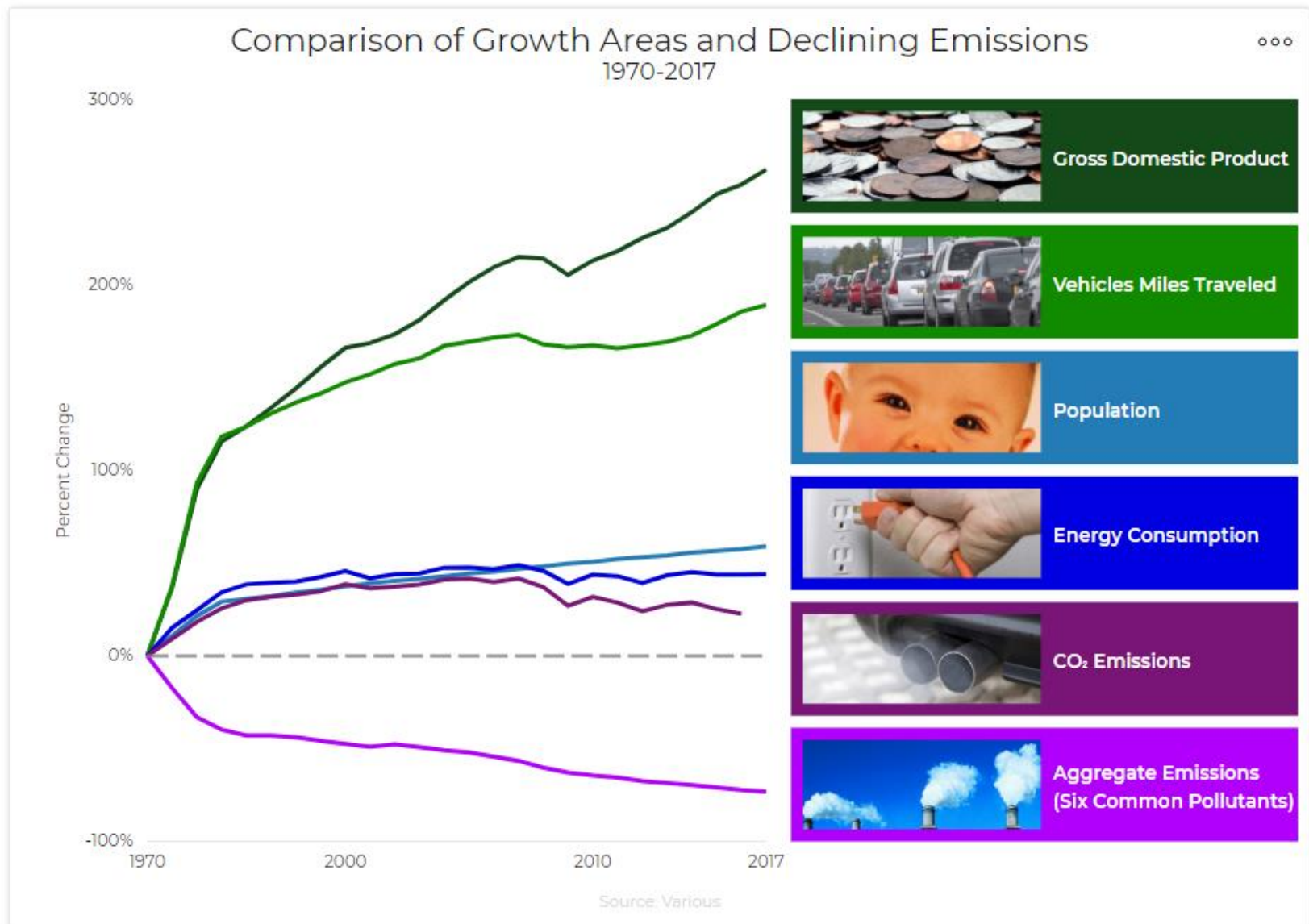
- 6 Criteria Air Pollutants (CAPs):
 - O₃, CO, NO₂, SO₂, Pb, PM (PM_{2.5} & PM₁₀)
 - Concentration level set in NAAQS (in CAA)
 - Ubiquitous, most harmful to health and environment
- 187 Hazardous Air Pollutants (HAPs)
 - A.K.A. toxic air pollutants or air toxics
 - Known/suspected to cause cancer, or other serious health or environmental effects
 - Goal: reduce emissions (NESHAPS)
 - Primary, man-made emissions dominates

Concentration trend in the US



Source: U.S. EPA Air Quality System

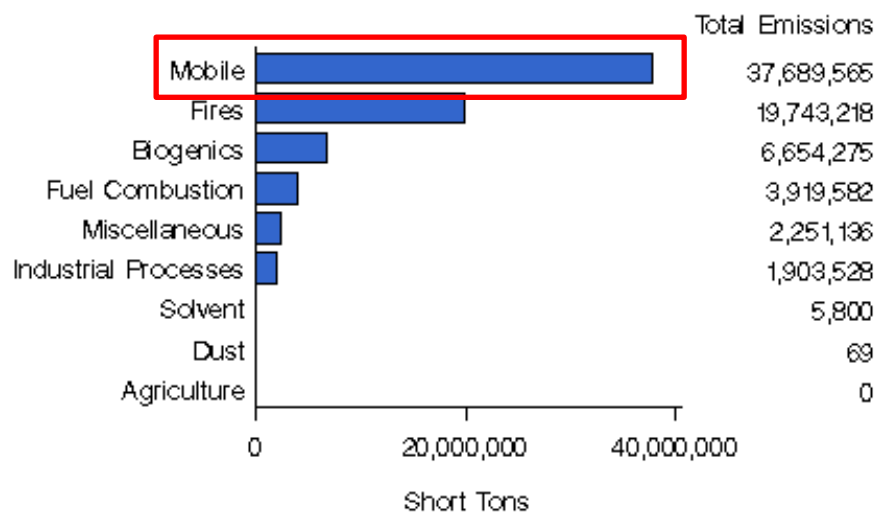
Emission trend in the US



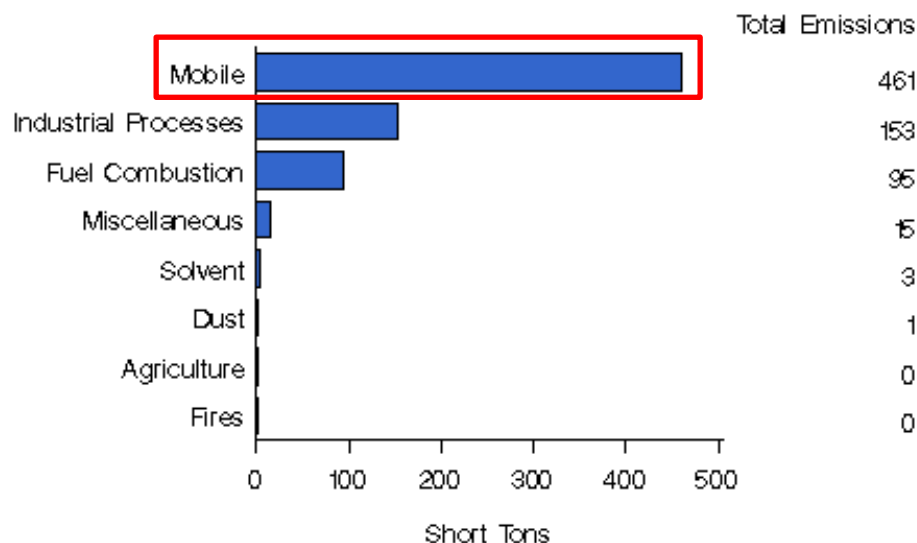
Emission sources

- Primary vs secondary
 - Directly emitted vs formed in the air
- Natural vs man-made (anthropogenic)
- Stationary vs mobile
 - Stationary stacks, moving cars etc...
- And also by shapes
 - Point, nonpoint, area, volume, line etc....

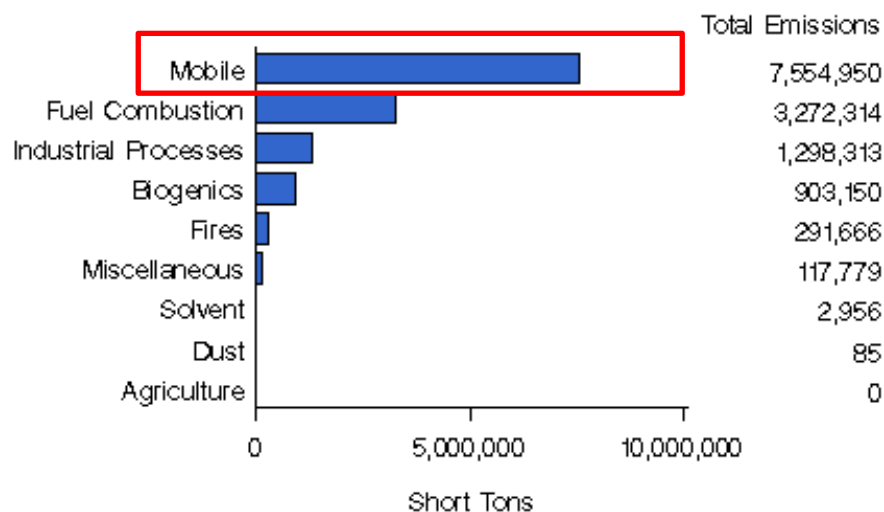
National Carbon Monoxide Emissions by Source Sector
(NEI 2014 v1)



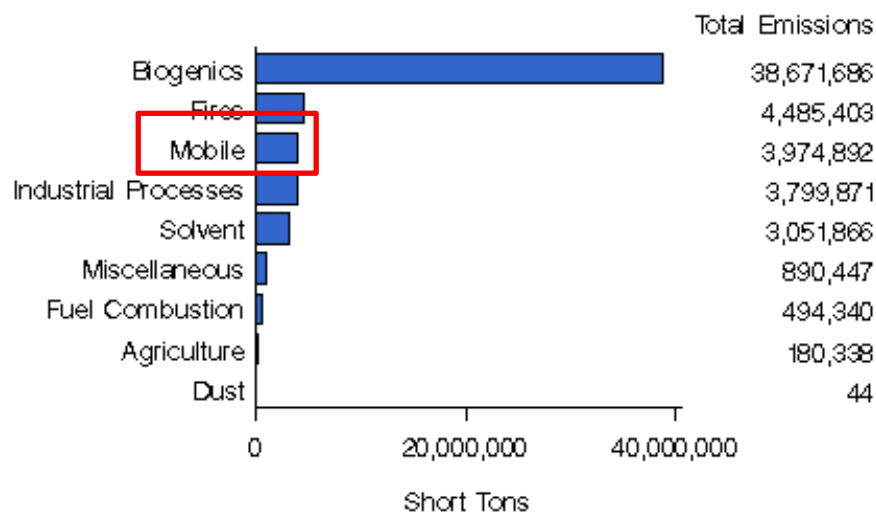
National Lead Emissions by Source Sector
(NEI 2014 v1)



National Nitrogen Oxides Emissions by Source Sector
(NEI 2014 v1)

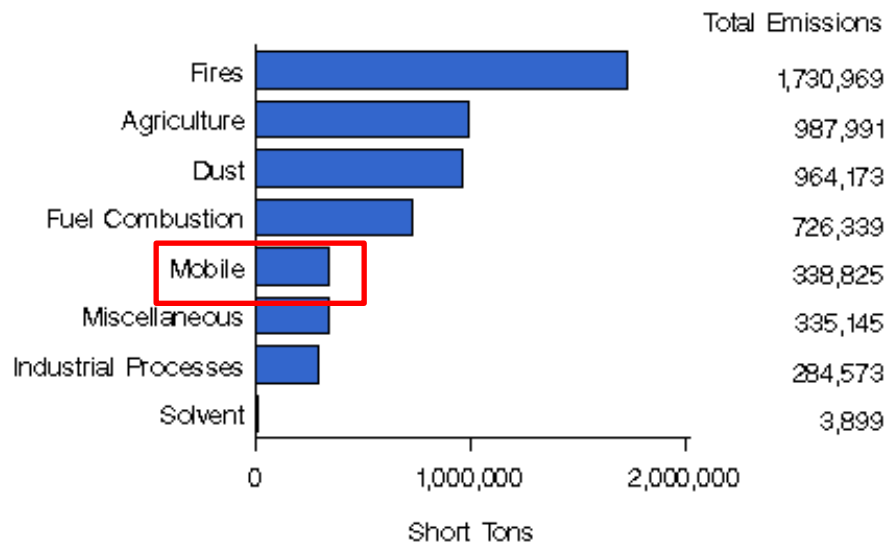


National Volatile Organic Compounds Emissions by Source Sector
(NEI 2014 v1)

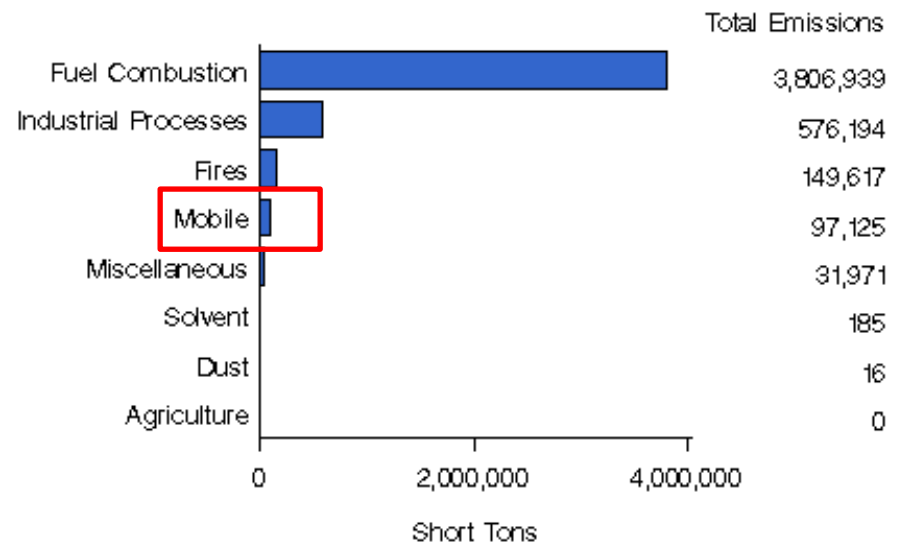


Source: USEPA (<https://www.epa.gov/air-emissions-inventories/air-emissions-sources>)

National PM_{2.5} Emissions by Source Sector
(NEI 2014 v1)



National Sulfur Dioxide Emissions by Source Sector
(NEI 2014 v1)



Source: USEPA (<https://www.epa.gov/air-emissions-inventories/air-emissions-sources>)

How mobile emissions were calculated

$$\mathbf{E = A * F * adj}$$

E: emissions (e.g. lb, lb/hr);

A: activity (mileage driven,...)

F: emission factor (amount of pol/activity)

adj: adjustment factor